Household Search, Wages and Commuting Time (Preliminary and Incomplete)

Marcela Perticará[1] Mauricio Tejada[2]

[1]Universidad Diego Portales

[2]Universidad Diego Portales

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- A job attribute that has received limited attention as a contributor to gender wage gaps is the strong preference for shorter commutes.
  - In OECD countries: 33 vs. 22 minutes on average. Most of the gap is suppy-side driven [Le Barbanchon et al., 2021]
  - Commuting time significantly matters for job satisfaction and subjective well-being [Clark et al., 2020]
  - Existing empirical papers have found a positive and robust relationship between commuting and wages [Petrongolo and Ronchi, 2020].

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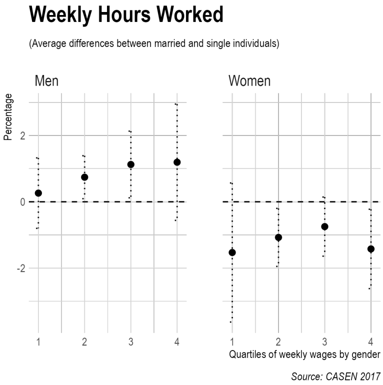
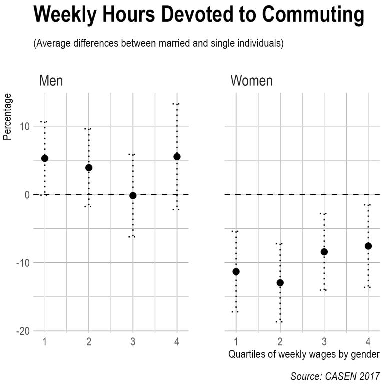
  - Commuting time significantly matters for job satisfaction and subjective well-being [Clark et al., 2020]
  - Existing empirical papers have found a positive and robust relationship between commuting and wages [Petrongolo and Ronchi, 2020].
- Household-level decisions play a significant role in shaping an individual's job search [Flabbi and Mabli, 2018].
  - Employment status and caregiving responsibilities of household members can impact when, where, and how a job is searched.

In Chile (CASE 2017), married women tend to work fewer hours than single women, but the main margin of adjustment seems to be the time devoted to commuting.





Questions:

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What do we do:

- We develop a household search model in which a job has two attributes: wage and commuting time.
  - Decisions to participate in the labor market, accept a job, and continue employment are endogenous.
- We structurally estimate the model using data from Chile.
  - Chile has one of the highest commuting times among the OECD countries (60 mins/day vs. 35 mins/day). [See OECD Family Database]
- We use counterfactual experiments to disentangle the role of job offers, labor market dynamics, and preferences on the MWP.

Why:

- Measuring WTP is essential for any cost-benefit analysis: it quantifies the monetary value of time; however, it is empirically challenging.
    - Hedonic wage regressions generate biased estimates (endogeneity, unobservables, etc.).
    - Discrete choice models of travel behavior have an identification problem (value of time vs. value of attributes of traveling).
    - Exploiting duration data (exit rates to obtain better attributes) works only under symmetric search environments.
    - Spatial search models (closest opportunity model) work very well but only with distance data.
- While the theoretical significance of considering the household as the decision-making unit is gaining acceptance/understanding, there is limited knowledge about its empirical impact on labor market outcomes. [Dey and Flinn, 2008].

- **Household job search**. Contribution: Commuting Time

  [Dey and Flinn, 2008, Guler et al., 2012, Mankart and Oikonomou, 2017, Flabbi and Mabli, 2018, García-Pérez and Rendon, 2020, Flinn et al., 2023, Pilossoph and Wee, 2021, Salazar-Saez, 2023]

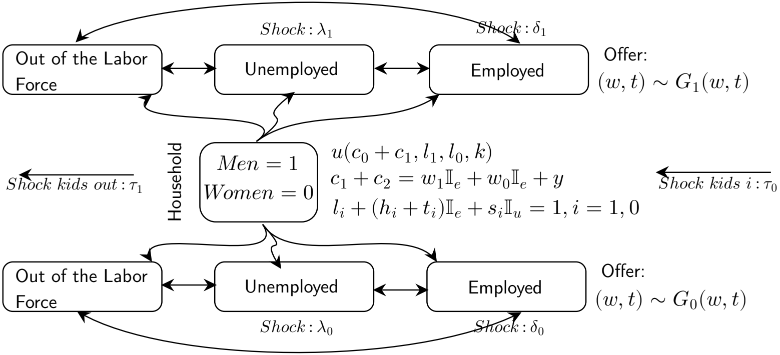
- **Job search and commuting time**. Contribution: Household Search

  [van den Berg and Gorter, 1997, Van Ommeren, 2018, Carra et al., 2016, van Ommeren et al., 2000, Rupert et al., 2009, Flemming, 2020, Le Barbanchon et al., 2021]

- **Structural estimation of job search models**. Contribution: Joint model of household search and Commuting Time

  [Flinn and Heckman, 1982, Eckstein and van den Berg, 2007, Dey and Flinn, 2008, Flabbi and Mabli, 2018, Flinn et al., 2023, Salazar-Saez, 2023]

Outline



Lifetime utilities:

$$U(k),\ T_i(w_i, t_i, k),\ V(w_1, w_0, t_1, t_0, k),\ R(k),\ W_i(w_i, t_i, k),\ Z(k);\ i = 1, 0$$

$$MWP = \frac{\partial[lifetime\ utility]}{\partial t} \Bigg/ \frac{\partial[lifetime\ utility]}{\partial t}$$

Model Details

- The data available is: CASEN 2017:
  - Type of worker: $\{I_1, I_0\}$ (couples)
  - Kids (under 12 years old) at home: $\{K\}$
  - Weekly wages for husband and wife (if employed): $\{w_1, w_0\}$.
  - Weekly commuting time for husband and wife (if employed): $\{t_1, t_0\}$.
  - States in the labor market: $\{I_{1,np=1}, I_{1,u=1}, I_{1,e=1}, I_{0,np=1}, I_{0,u=1}, I_{0,e=1}\}$.
  - Ongoing employment and unemployment durations (if employed or unemployed, respectively): $\{t_1^u, t_1^e, t_0^u, t_0^e\}$.
  - Searching time (National Time-Use Survey): $\{s_1, s_0\}$

- The data available is: CASEN 2017:
  - Type of worker: $\{l_1, l_0\}$ (couples)
  - Kids (under 12 years old) at home: $\{K\}$
  - Weekly wages for husband and wife (if employed): $\{w_1, w_0\}$.
  - Weekly commuting time for husband and wife (if employed): $\{t_1, t_0\}$.
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- Sample:
  - Heterosexual couples (married/cohabitating), individuals aged between 25 and 55 years living in urban areas. For those employed, we use full-time equivalent employees.

	Males			Females		
Spouse Labor Market	NP	U	E	NP	U	E
Labor market state						
NP	0.017	0.001	0.020	0.017	0.014	0.424
U	0.014	0.006	0.022	0.001	0.006	0.037
E	0.424	0.037	0.459	0.020	0.022	0.459
Presence of kids						
Mean	0.625	0.578	0.568	0.439	0.551	0.603
Weekly Wages (US$)						
Mean	216.17	230.03	266.69	181.33	178.25	227.71
SD	183.99	185.22	224.51	143.85	138.36	191.27
Weekly Commuting Time (Hours)						
Mean	6.651	6.865	6.120	5.452	5.233	4.856
SD	5.518	5.523	5.065	4.946	4.285	4.152
Corr with wages	0.032	-0.093	-0.017	0.047	0.014	0.022
Duration (Months)						
Mean (E)	103.384	83.819	95.681	85.067	73.017	77.120
Mean (U)	2.464	3.280	2.626	1.413	2.548	2.139

NOTE: $N = 9273$ households. SD = Standard deviation; E = employed full time; U = unemployed.

Estimation:

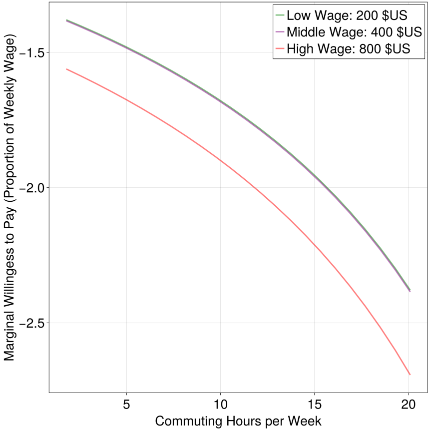
- We estimate the primitive parameters of the model using the Method of Simulated Moments (39 moment statistics to estimate 20 parameters).
  - Today: Only labor market participants.

Identification:

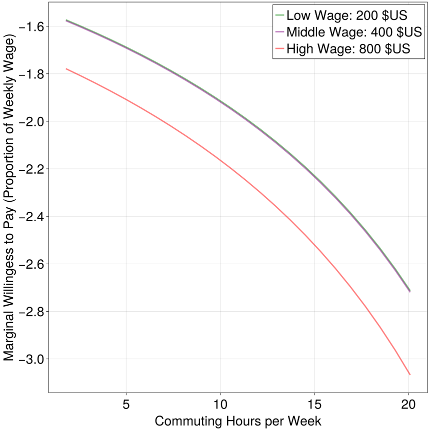
- **The (wage, commuting time) offer distribution**: bivariate log-normal distribution + accepted wages and commuting time information (recoverability in the conditional marginals).
- **Model dynamics (arrival rates)**: steady-state proportion of workers in each labor market state + the duration information.
- **Preferences**: CRRA + unitary model + wages info on consumption (hand-to-mouth households) + commuting times info preferences for leisure.

See Identification Details

Figure: Wife's Marginal Willingness to Pay when Husband is Unemployed

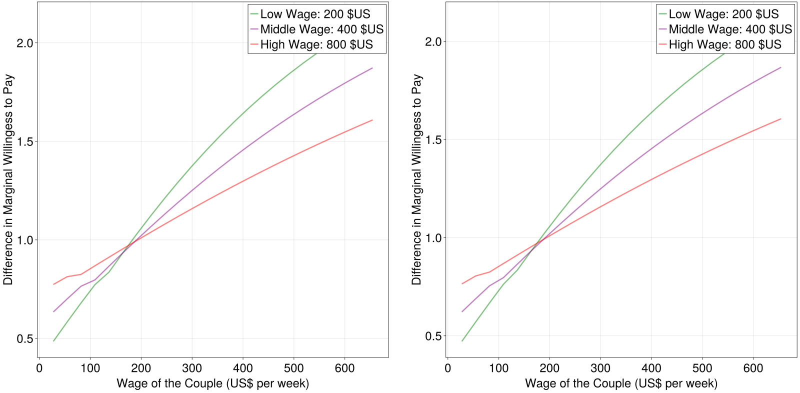


(a) No Kids



(b) Kids

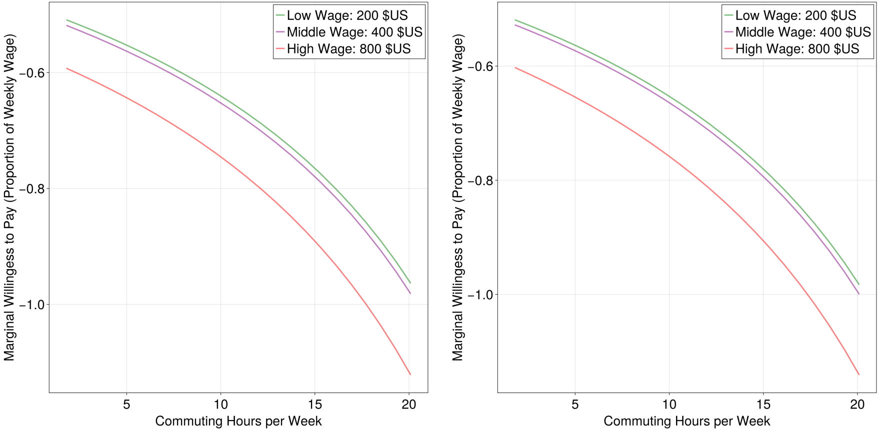
Figure: Wife's Marginal Willingness to Pay: Husband Employed vs. Husband Unemployed



(a) No Kids (b) Kids

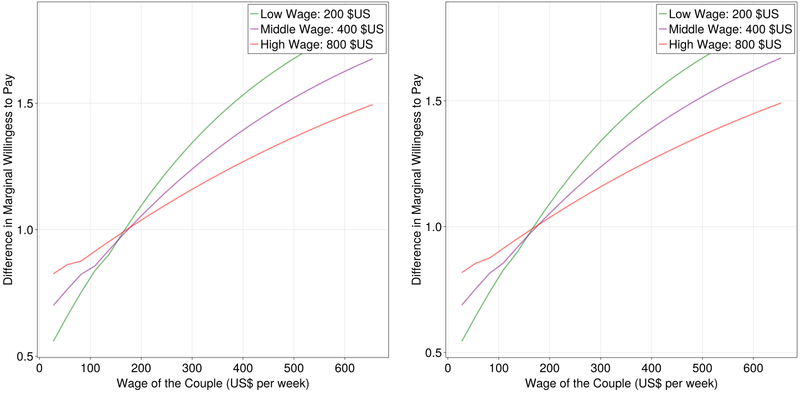
$MWP^{\text{Husband Employed}} / MWP^{\text{Husband Unemployed}}$ as function of husband's wage

Figure: Husband's Marginal Willingness to Pay when Wife is Unemployed



(a) No Kids (b) Kids

Figure: Husband's Marginal Willingness to Pay: Wife Employed vs. Wife Unemployed



(a) No Kids (b) Kids

$MWP^{\text{Wife Employed}}/MWP^{\text{Wife Unemployed}}$ as function of wife's wage

We have three sources of differences between husbands and wives:

- Utility over leisure (non labor market activities) $\rightarrow$ Gender differences interpretation: *Valuation of home production and caregiving responsibilities*
- Job offer distributions $\rightarrow$ Gender differences interpretation: *Availability of information, types of jobs being searched for, discrimination*
- Labor market dynamics $\rightarrow$ Gender differences interpretation: *Intensity of search, job instability, discrimination*

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We perform two counterfactual simulations (turning off sources of gaps):

- Equal Offer Distributions: $G_0(w_0, t_0) = G_1(w_1, t_1)$.
- Equal Offer Distributions + Equal Labor Market Dynamics: $G_0(w_0, t_0) = G_1(w_1, t_1)$, $\lambda_1 = \lambda_0$, and $\delta_1 = \delta_0$.

	Benchmark	Same Offer Distributions	Same Offer Distributions and L.M. Dynamics
Average Wages (US$ per Week)			
Men with couple employed	275	268	290
Men with couple unemployed	241	233	268
Women with couple employed	235	286	292
Women with couple unemployed	206	257	256
Average Commuting Hours per Week			
Men with couple employed	6.3	6.4	6.3
Men with couple unemployed	6.0	6.0	6.5
Women with couple employed	5.0	5.9	6.0
Women with couple unemployed	4.9	6.0	5.8
Ratio (Men/Women)			
Wages - Couple employed	1.17	0.94	1.00
Wages - Couple unemployed	1.17	0.91	1.04
Commuting - Couple employed	1.26	1.08	1.05
Commuting - Couple unemployed	1.23	1.00	1.12

Main Results:

- The MWP for women is considerably larger than the MWP for men. The presence of children increases the disutility of commuting.
- MWP increases with both one's own wage and one's partner's wage.
- Differences in offer distributions and preferences explain the gaps in commuting time. This also partially explains wage gaps.

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What is missing:

- Heterogeneity: Analyze differences by education level (with sorting: skill-skill and unskilled-unskilled).
- Mono-parental and extended families.
- Moving decisions (challenging because of lack of information).
- Teleworking (declared $t = 0$ means telework?)

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- Time is continuous, individuals leave forever and the future is discounted by ρ.
- Two type of workers: men indexed by 1 and women indexed by 0.
- Labor market states: Each individual can be out of the labor force, unemployed or employed.
- A job offer arrives at Poisson rate λ. Full-time jobs.
- Individuals spend h working in full-time jobs.
- A job offer is a vector (w, t), where the wage w and the commuting time (in hours) t which are draws of the joint distributions $G(w, t)$ for full-time.
- Jobs are terminated exogenously at Poisson rates δ.
- When unemployed, individuals spend s hours searching for a job (cost of searching).
- Households without kids at home receive a kid shock at Poisson rate τ_0. Households with kids are relieved from the kid responsibility at Poisson rate τ_1.

- Unitary model: The household solves the following lifetime problem by choosing the path of each member in the labor market:

$$\max \quad \int_0^\infty u(c_1 + c_0, l_1, l_0) e^{-\rho t} dt \tag{1}$$

subject to

$$c_1 + c_0 = w_1 \mathbb{I}_1^e + w_0 \mathbb{I}_0^e + y$$
$$l_1 = 1 - (h_1 + t_1)\,\mathbb{I}_1^e - s_1 \mathbb{I}_1^u$$
$$l_0 = 1 - (h_0 + t_0)\,\mathbb{I}_0^e - s_0 \mathbb{I}_0^u$$

- $\mathbb{I}$ are indicator variables that take the value of 1 if employed (e) or unemployed (u), and y non labor market income.
- Household members engage in a leader-follower game when making decisions; there is no bargaining involved in the decision-making process.

Definitions of lifetime values:

- Both members of the household are working: $V(w_1, t_1, w_0, t_0, s_1, s_0, k)$.
- One member of the household is working and the other unemployed: $T_i(w_i, t_i, s_1, s_0, k)$ with $i = 0, 1$.
- One member of the household is working and the other is out of the labor force: $W_i(w_i, t_i, s_1, s_0, k)$ with $i = 0, 1$.
- Both members of the household are unemployed: $U(s_1, s_0, k)$.
- One member of the household is unemployed and the other is out of the labor force: $R_i(s_1, s_0, k)$ with $i = 0, 1$.
- Both members of the household are out of the labor force: $Z(s_1, s_0, k)$.

Both members of the household are employed:

$$\begin{aligned}
\rho V(w_1, t_1, w_0, t_0, s_1, s_0, k) &= u(w_1 + w_0 + y, 1 - h_1 - t_1, 1 - h_0 - t_0, k) \\
&+ \delta_0 Q_1(w_1, t_1, s_1, s_0, k) + \delta_1 Q_0(w_0, t_0, s_1, s_0, k) \\
&+ [\tau_1 k + \tau_0(1-k)] \max \left\{ \begin{array}{l} V(w_1, t_1, w_0, t_0, s_1, s_0, 1-k), \\ Q_1(w_1, t_1, s_1, s_0, 1-k), \\ Q_0(w_0, t_0, s_1, s_0, 1-k) \end{array} \right\} \\
&- (\delta_0 + \delta_1 + \tau_1 k + \tau_0(1-k)) V(w_1, t_1, w_0, t_0, s_1, s_0, k)
\end{aligned}$$

where

$$Q_i(w_i, t_i, s_1, s_0, k) = \max \{ T_i(w_i, t_i, s_1, s_0, k), W_i(w_i, t_i, s_1, s_0, k) \} \quad i = 1, 0$$

One member of the household is employed and the other is unemployed:

$$
\begin{aligned}
\rho T_i(w_i, t_i, s_1, s_0, k) = {} & i \times u(w_1 + y, 1 - h_1 - t_1, 1 - s_0, k) \\
& + (1 - i) \times u(w_0 + y, 1 - s_1, 1 - h_0 - t_0, k) + \delta_1 S(s_1, s_0, k) \\
& + \lambda_0 \int \int \max \left\{ \begin{array}{l} T_i(w_i, t_i, s_1, s_0, k), \\ V(w_1, t_1, w_0, t_0, s_1, s_0, k), \\ Q_{1-i}(w_{1-i}, t_{1-i}, s_1, s_0, k) \end{array} \right\} g_0(w_0, t_0) dw_0 dt_0 \\
& + [\tau_1 k + \tau_0(1 - k)] \max \{ T_i(w_i, t_i, s_1, s_0, 1 - k), S(s_1, s_0, 1 - k) \} \\
& - (\delta_1 + \lambda_0 + \tau_1 k + \tau_0(1 - k)) \, T_i(w_i, t_i, s_1, s_0, k); \quad i = 1, 0
\end{aligned}
$$

where

$$
Q_i(w_i, t_i, s_1, s_0, k) = \max \{ T_i(w_i, t_i, s_1, s_0, k), W_i(w_i, t_i, s_1, s_0, k) \}
$$

and

$$
S(s_1, s_0, k) = \max \{ U(s_1, s_0, k), R_1(s_1, s_0, k), R_0(s_1, s_0, k), Z(s_1, s_0, k) \}
$$

One member of the household is employed and the other is out of the labor force:

$$
\begin{aligned}
\rho W_i(w_i, t_i, s_1, s_0, k) &= i \times u(w_1 + y, 1 - h_1 - t_1, 1, k) \\
&+ (1 - i) \times u(w_0 + y, 1, 1 - h_0 - t_0, k) \\
&+ \delta_i \max\{R_i(s_1, s_0, k), U(s_1, s_0, k), Z(s_1, s_0, k)\} \\
&+ [\tau_1 k + \tau_0(1 - k)] \max\left\{W_i(w_1, t_1, s_1, s_0, 1 - k), S(s_1, s_0, k - 1)\right\} \\
&- (\delta_1 + \tau_1 k + \tau_0(1 - k))\, W_i(w_1, t_1, s_1, s_0, k); \quad i = 1, 0
\end{aligned}
$$

The optimal decision rules in this model have a reservation value property:

- $w_i^*(t_i) \rightarrow U = T_i(w_i^*(t_i), t_i)$ with $i = 1, 0$
- $\tilde{w}_0^*(w_1, t_1, t_0) \rightarrow T_1(w_1, t_1) = V(w_1, t_1, \tilde{w}_0^*(w_1, t_1, t_0), t_0)$ for women
- $\tilde{w}_1^*(t_1, w_0, t_0) \rightarrow T_0(w_0, t_0) = V(\tilde{w}_1^*(t_1, w_0, t_0), t_1, w_0, t_0)$ for men.

Definition

Given $\{\lambda_i, \delta_i, \beta_i, \alpha_i\}_{i=1,0} \times \{\gamma, y, \rho\}$ and $G_i(w_i, t_i)$ with $i = 1, 0$, a steady-state equilibrium is a set of value functions $\{V(\cdot), T_1(\cdot), T_0(\cdot), W_1(\cdot), W_0(\cdot), U, Z\}$; the reservation values $w_i^*(t_i)$ and $\tilde{w}_i^*(w_{1-i}, t_{1-i}, t_i)$ with $i = 1, 0$; and the proportions of households with both members employed, both members unemployed, and one member employed while the other is unemployed. These satisfy the Bellman equations and maintain an invariant distribution of labor market states across individuals and households.

Return

- The Marginal Willingness to Pay (MWP) is to the extra amount of money that the worker is willing to pay for a job attribute (in our case: lower time of commute).

$$MWP = \frac{\partial \text{E[Lifetime Utility]}}{\partial t} \Big/ \frac{\partial \text{E[Lifetime Utility]}}{\partial w}$$

- The MWP compares the relative importance of the current attributes of w and t for the expected lifetime utility.
- What is relevant for our paper is that the MWP depends not only on the person's position in the labor market but also on the labor market position of the household members.

$$\begin{aligned} MWP_j^{couple\ \ u}(w_j, t_j) &= \frac{\partial T_j(w_j, t_j)}{\partial t_j} \Big/ \frac{\partial T_j(w_j, t_j)}{\partial w_j} \\ MWP_j^{couple\ \ e}(w_j, t_j, w_{-j}, t_{-j}) &= \frac{\partial V(w_j, t_j, w_{-j}, t_{-j})}{\partial t_j} \Big/ \frac{\partial V_j(w_j, t_j, w_{-j}, t_{-j})}{\partial w_j} \end{aligned}$$

for $j = 1, 0$.

The endogenous decision of moving close to the working place:

$$\begin{aligned}\rho T_1(w_1, t_1) = {} & u(w_1 + y, 1 - h_1 - t_1, 1 - s_0) + \delta_1 \left[U - T_1(w_1, t_1)\right] \\ & + \lambda_0 \int\int \tilde{T}_1(w_1, t_1, w_0, t_0) g_0(w_0, t_0) dw_0 dt_0 \\ & - \lambda_0 T_1(w_1, t_1) \end{aligned} \tag{2}$$

$$\begin{aligned}\tilde{T}_1(w_1, t_1, w_0, t_0) = \max \{ & T_1(w_1, t_1), V(w_1, t_1, w_0, t_0), T_0(w_0, t_0), \\ & V(w_1, \tilde{t}_1(t_0), w_0 - c, 0), T_0(w_0 - c, 0)\}\end{aligned}$$

$$\tilde{t}_1(t_0) = p \max(0, t_1 - t_0) + (1 - p) \min(t_1 + t_0, 1 - h_1)$$

$$\begin{aligned}\rho U = u(y, 1 - s_1, 1 - s_0) + \lambda_1 \int\int \max\{U, T_1(w_1, t_1), T_1(w_1 - c, 0)\} g_1(w_1, t_1) dw_1 dt_1 \\ + \lambda_0 \int\int \max\{U, T_0(w_0, t_0), T_0(w_0 - c, 0)\} g_0(w_0, t_0) dw_0 dt_0 - (\lambda_1 + \lambda_0) U \end{aligned} \tag{3}$$

- The (wage, commuting time) offer distribution parameters are mainly identified from the accepted wages and commuting time information.
  - Data for accepted jobs: truncation of offer distributions.
  - We assume a bivariate log-normal distribution:

$$\begin{bmatrix} \log w \\ \log t \end{bmatrix} \sim \mathcal{N}\left(\begin{bmatrix} \mu_w \\ \mu_t \end{bmatrix}, \begin{bmatrix} \sigma_w^2 & \sigma_{wt} \\ \sigma_{wt} & \sigma_t^2 \end{bmatrix} \right)$$

- Flinn and Heckman [1982]: Recoverability condition (primitive can be revered from a truncated version). Lognormal in one dimension satisfies this condition.
- Given the reservation values property we only need:

$$\log w \mid \log t \sim \mathcal{N}\left(\mu_{w|t}, \sigma^2_{w|t}\right)$$

where $\mu_{w|t} = \mu_w + \rho_{w,t}\frac{\sigma_w}{\sigma_t}\left(t - \mu_t\right)$ and $\sigma^2_{w|t} = \sigma^2_w\left(1 - \rho^2_{w,t}\right)$

- We assume a constant relative risk aversion (CRRA) formulation.

$$u\left(c, l_1, l_0, k; \gamma, \beta, \alpha\right) = \left(1 - \alpha_1 - \alpha_0(k)\right) \frac{c^{\gamma} - 1}{\gamma} + \alpha_1 \frac{l_i^{\beta_1} - 1}{\beta_1} + \alpha_0(k) \frac{l_j^{\beta_0} - 1}{\beta_0}$$

with $\alpha_0(k) = \alpha_0 + \phi \times k$

  - Guler et al. [2012] and Flabbi and Mabli [2018]: Risk aversion matters for decisions in a unitary household model.
  - Accepted wages provide information about consumption (given hand-to-mouth households) and differences in accepted commuting times by husbands and wives provide information on preferences for leisure.
- Flinn and Heckman [1982]: The mobility parameters are mainly identified by the steady-state proportion of workers in each labor market state and by the duration information.

Return

	Males		Females	
	Parameter	Std.Error	Parameter	Std.Error
λ	0.3538	(0.01497)	0.3747	(0.03646)
δ	0.0102	(0.01755)	0.0138	(0.0883)
μ_w	0.7065	(0.06641)	0.4729	(0.06244)
μ_t	-2.7460	(0.02201)	-2.9994	(0.0065)
σ_w	0.8388	(0.20160)	0.8454	(0.4658)
σ_t	0.7158	(0.03273)	0.7619	(0.00996)
$\rho_{w,t}$	-0.0060	(0.04654)	0.0316	(0.08838)
β	0.8485	(0.01647)	0.7244	(0.03674)
α	0.0420	(0.05055)	0.1124	(0.07423)
	Household			
γ	1.6857	(0.9512)		
ϕ	0.0217	(0.0435)		
y	0.00006	(0.6725)		
Loss		1515		

NOTE: Bootstrapped standard errors in parenthesis. Fixed parameters $\rho = 0.1$, $\tau_0 = 0.0877$, prop of households with kids 0.594, $h_1 = h_0 = 44/80$, $s_1 = 11/80$, and $s_0 = 11/80$.